

Learning Guide Unit 8

Overview

UNIT 8: Exploring Graphs and Advanced Graph Algorithms

Topics

- Introduction to Graphs: Representation (Adjacency Matrix, Adjacency List).
 - Graph Algorithms: Depth First Search (DFS), Breadth First Search (BFS), Dijkstra's Algorithm (Shortest Path).
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Learning Objectives

By the end of this Unit, you will be able to:

1. Identify graph representations (adjacency matrix, adjacency list) and their applications in algorithms.
2. Implement Depth First Search (DFS) and Breadth First Search (BFS) algorithms to explore graph structures.
3. Apply Dijkstra's algorithm to solve shortest path problems.